

EDUCATION

University of California, Irvine

Master of Computer Science

Irvine, CA

December 2027

Coursework - Algorithms, Big Data Systems, Operating Systems, Artificial Intelligence, Advanced Programming

Indian Institute of Technology, Kharagpur (IIT KGP)

Dual Degree (B.Tech (Hons)+ M.Tech) in Computer Science and Engineering

Kharagpur, India

July 2025

EXPERIENCE

Texas Instruments

July 2025 – July 2026

Software Engineer, Edge AI – Full-Time

- Designed and implemented 7 low-level-neural-network operator kernels — Add, Sub, Mul, Conv2D, Linear, AvgPool2D, Sigmoid — for a custom **Neural Processing Unit (NPU)**, optimizing **tiling**, **DMA scheduling**, **memory-bank access** and **schedule generation** to support downstream model deployment.
- Implemented N-dimensional broadcasting for Add/Sub/Mul via **compact DMA** and **vector-SRAM replay**, enabling complete coverage across **8** broadcast categories and broader operator support.
- Built a reusable deep-learning test framework validating kernels against PyTorch reference across **10,000+** test cases and **5** numeric precisions, diagnosing accelerator defects including **OOB DMA**, **memory corruption**, **NaN propagation**, and **BF16 drift** ahead of hardware bring-up.
- Built a **Claude skill** to auto-generate kernel code from finalized design specifications, cutting new-operator development time by **~4×** and accelerating the node-library roadmap.
- Deployed production object-detection models to INT8 via **quantization-aware distillation** and an **ONNX-to-PyTorch pipeline**, retaining **97–100% FP32 accuracy** for edge deployment.

Texas Instruments

May 2024 – July 2024

Embedded Software Intern, Sitara Microprocessor Unit Team (MPU)

- Redesigned Git PR static-analysis workflow using **Jenkins** and **Klocwork**, replacing nightly builds with PR-triggered checks and reducing analysis time by **6×** through selective modified-file scanning.
- Developed **Shell scripts** for automating coding-standard checks on modified files, enforcing **MISRA C/C++ compliance** and enabling end-to-end static analysis within the CI/CD workflow.
- Explored dynamic analysis with **LDRA**, identifying a CLI approach to eliminate GUI dependency and enable a touch-free developer workflow, with automation scoped for integration beyond the scope of my internship.

SKILLS

Development – Typescript | ReactJS | NextJS | REST API | Web Sockets | SSE | GraphQL | Claude Code | Agent Skills
Backend/Data – C++ | C | Python | Java | NodeJS | MySQL | MongoDB | Vector DB | Spark | Kafka | Hadoop | CI/CD | Git
AI/ML – PyTorch | TensorFlow | Keras | Transformers | OpenCV | ONNX | Quantization | VLM | Knowledge Distillation
Systems – Embedded Linux | Computer Architecture | x86-64 Assembly | Linux System Calls | Kernel Development

PROJECTS

Multimodal Benchmark for Vision-Language Reasoning on Rebus Puzzles

Aug 2024 – Apr 2025

M.Tech Thesis | Supervisor: Prof. Pawan Goyal | IIT Kharagpur [[arXiv:2511.01340](https://arxiv.org/abs/2511.01340)]

- Built a benchmark of 1,333 rebus puzzles across 18 categories to evaluate multimodal reasoning in **6** closed- and open-source **Vision Language Models (VLMs)**.
- Developed RebusDescProgICE combining code-based reasoning and embedding-based example selection, improving VLM performance by **2.1–4.1%** and **20–30%** on closed- and open-source models, respectively.

Caption-Enhanced Image Classification & Synthetic Data Augmentation

Aug 2023 – Apr 2024

B.Tech Thesis | Supervisor: Prof. Abir Das | IIT Kharagpur [[Paper](#)]

- Developed a multimodal classification framework combining **ViT** image embeddings with **BLIP-2/BERT** caption representations across CIFAR-100 and Caltech-101.
- Integrated **BLIP-2** and **Stable Diffusion** for caption generation and synthetic-data augmentation, achieving **96.5% Caltech-101** and **82.59% CIFAR-100 accuracy**.

RVL-CDIP - Image Classification

Aug 2022 – Nov 2022

Prof. Animesh Mukherjee and Prof. Saptarshi Ghosh | IIT Kharagpur [[Paper](#)]

- Implemented CNN, Transformer, LSTM, Few-Shot, and ensemble approaches for **16-class document classification**, achieving **77.0% test accuracy** with ensemble-by-loss.

AWARDS

National Talent Search Examination (NTSE) Scholar | 2018 - Selected as one of the top **~2,000 students nationally** out of **1.5M+** applicants in one of India's most competitive, government-funded talent recognition programs.